Back-Testing

Money managers will often construct portfolios via optimization processes. To help in the construction process these same managers will often run numerous studies testing different ideas, how prices react to different sets of factors, and how these portfolios perform under various market conditions. Interest lies not only in which companies will outperform or achieve excess returns, but rather how stocks will perform compared to different factors.

Quantitative managers spend much time and energy testing, re-testing, and verifying ideas and ensuring the uncovered relationships are statistically sound before settling on the preferred portfolio. Often this requires the investment ideas to be tested over a very long horizon, such as twenty to thirty years, or possibly more if data is available.

All too often, unfortunately, portfolio managers find a strategy works well in the back-testing analysis but once it is put into production the strategy does not achieve the expected level of return or worse case loses money. Why does this happen you ask. Well the primary reason is due to the implementation cost of the strategy. Costs are often much higher in reality than expected or planned. Of course, some of this is driven by the quant managers themselves. Quantitative managers do a great job at uncovering opportunities, but unfortunately all quants seem to find the same opportunities at the same time. This increases the buying and selling pressure in the stocks, leading to a higher cost. For example, if the manager's order is 5% ADV it is more likely that the group of quantitative managers will exert buying pressure in the stock close to possibly 50−75% ADV or more. This leads to higher trading costs in reality than would be reflective of the 5% order and could lead to reduced profits or losses. But it is also due to managers often using an unrealistic transaction cost estimate such as 20 bp or possibly just one half of the spread.

Another issue that could potentially arise is one where the manager deems the trading cost is too expensive and offsets the incremental alpha. Thus, the manager would abandon this strategy because they believe it would be unprofitable. However, many times managers are using historical costs that are too high or do not properly reflect today's trading environment. For example, it is possible that if we traded the same list historically based on today's market structure (decimals, algorithms, etc.) the transaction cost would be much lower than we expect thus making the strategy profitable. If the manager does not include a realistic trading cost expectation in their back-testing environment it may cause them to abandon a potentially profitable investment idea.